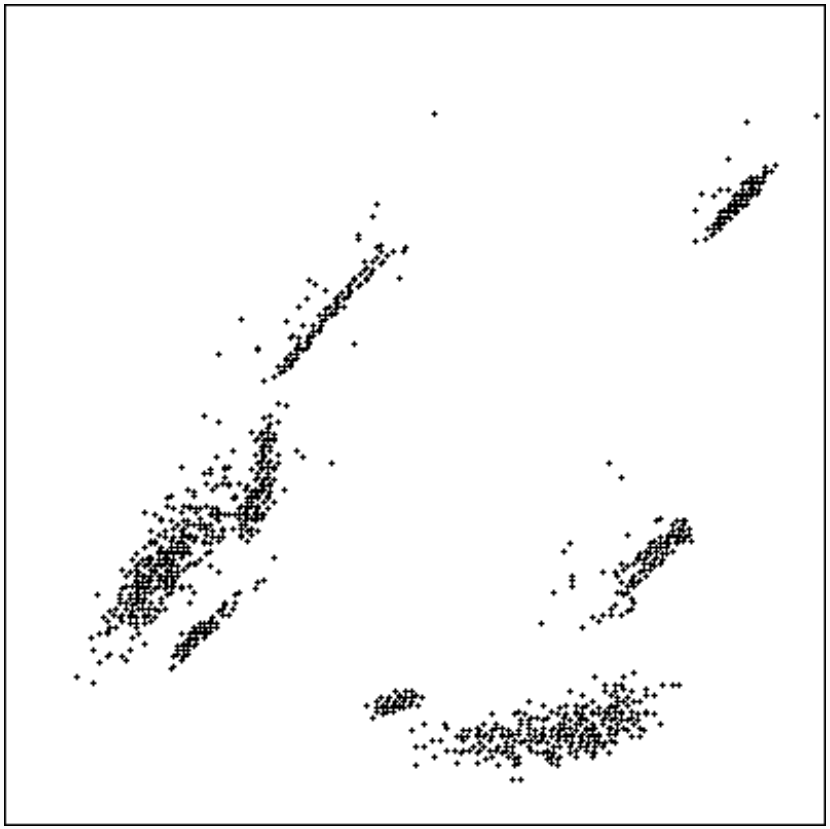
Clustering Techniques

Credit to: Pasi Fränti & Jing Gao

Sample Data

Sources of RGB vectors

Red-Green plot of the vectors



Sample Data

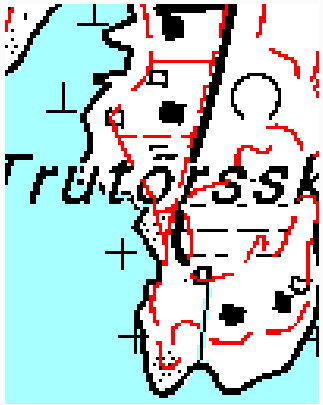
Employment statistics:

ID	POSTAL ZONE	Self employed	Civil servents	Clerks	Manual workers
800	Munchen	56750	57218	300201	242375
801	Munchen-land ost	7684	5790	20279	23491
802	Munchen-land sued	3780	1977	11058	7398
803	Munchen-land west	7226	5623	25571	20380
804	Munchen-land nord	2226	1305	9347	12432
805	Freising	8187	5140	14632	24377
806	Dachau	8165	2763	11638	24489
807	Ingolstadt	5810	5212	15019	30532
•	•	•	•	•	•
•	•	•	•	•	•

Application examples

Color Reconstruction

Image with original colors



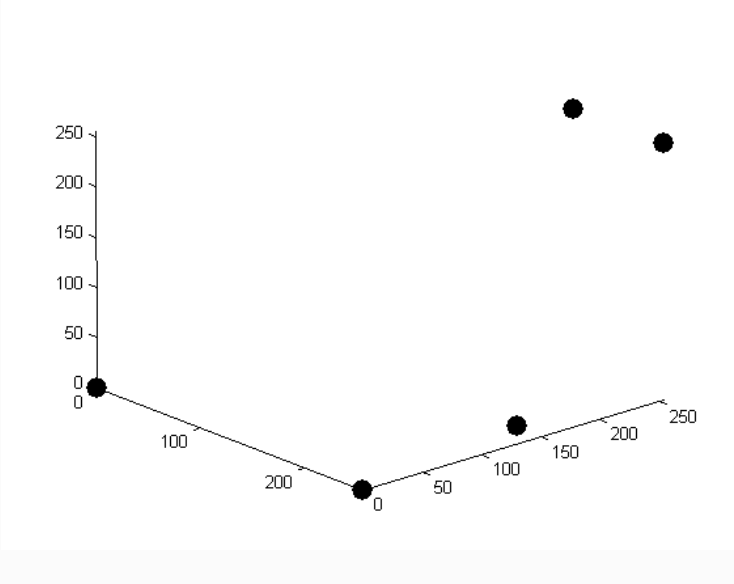
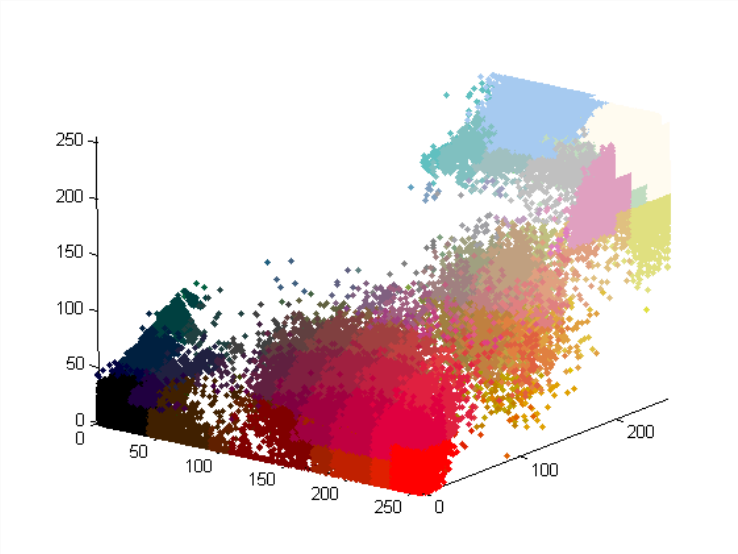
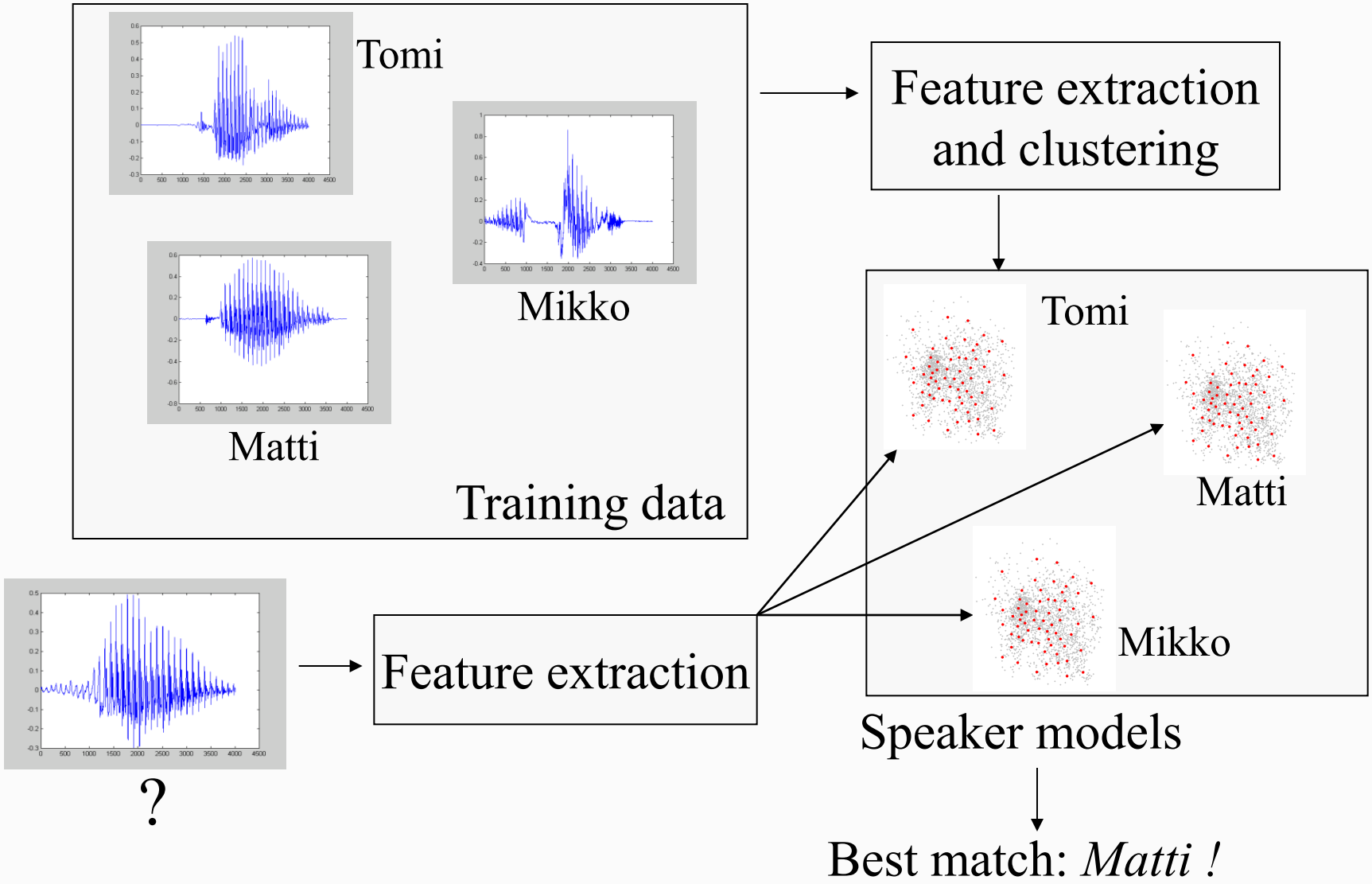


Image with compression artifacts

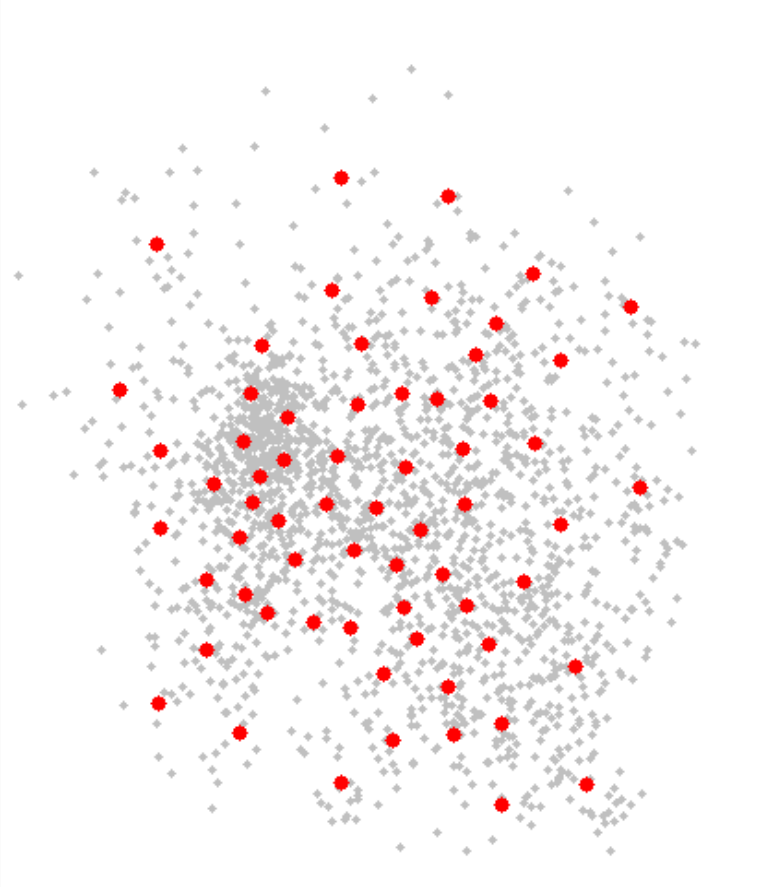


Speaker Modeling
for voice biometrics



Speaker Modeling

Speech data

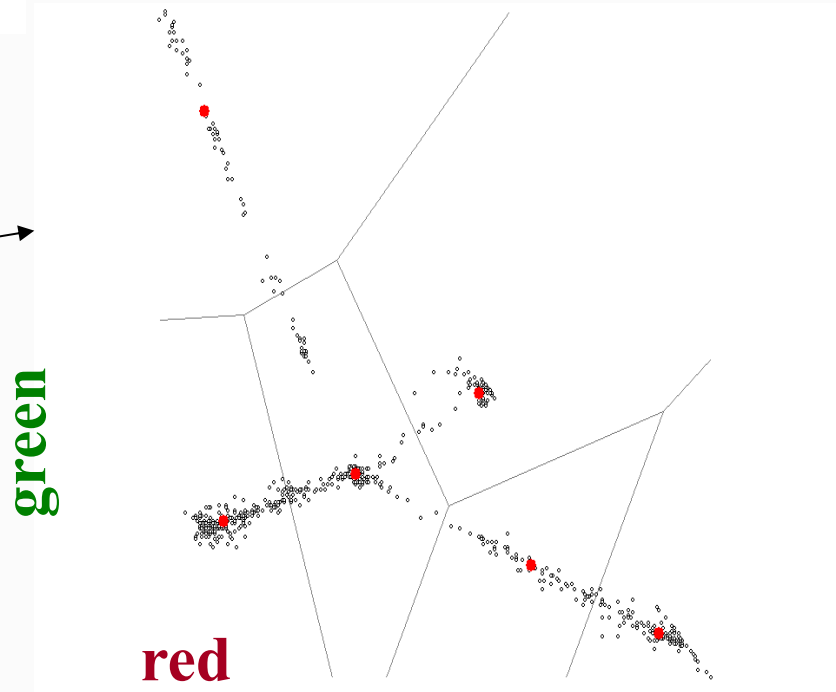


Result of clustering

Image segmentation

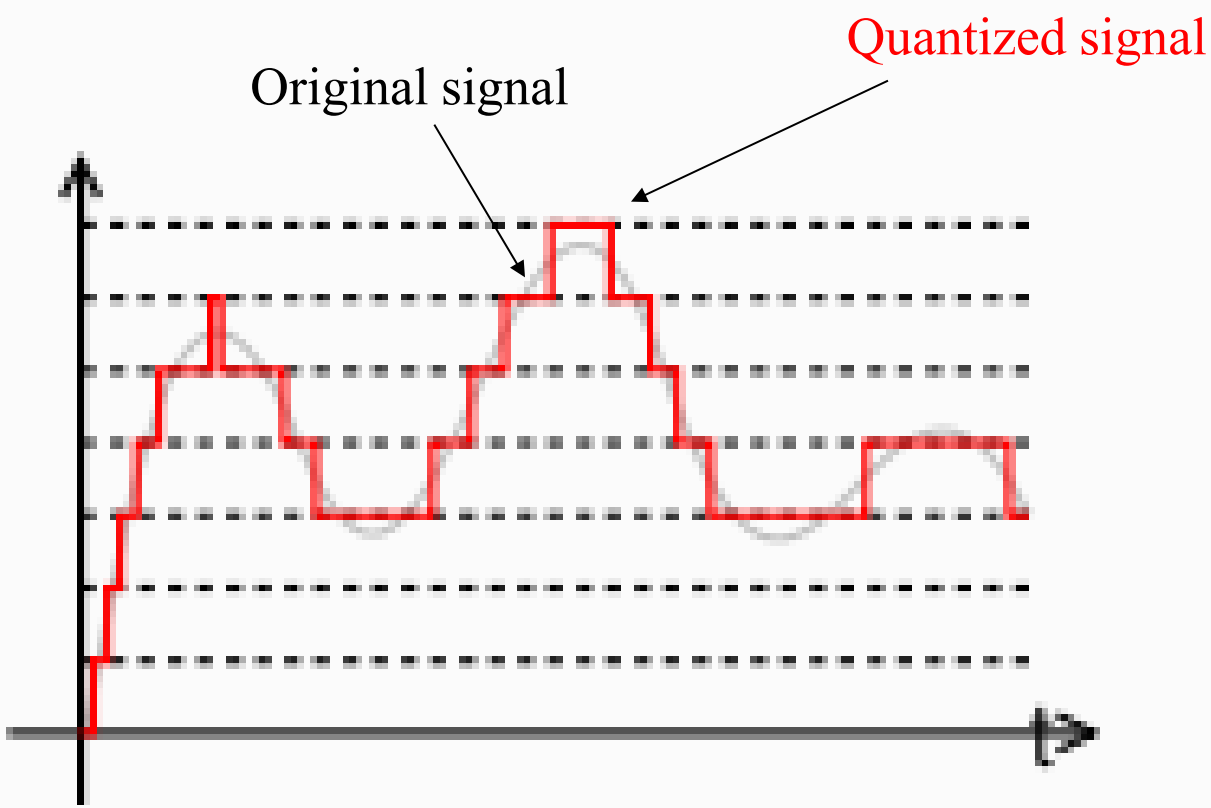
Image with 4 color clusters

Normalized color plots according to **red** and **green** components.



Signal Quantization

Approximation of continuous range values (or a very large set of possible discrete values) by a small set of discrete symbols or integer values



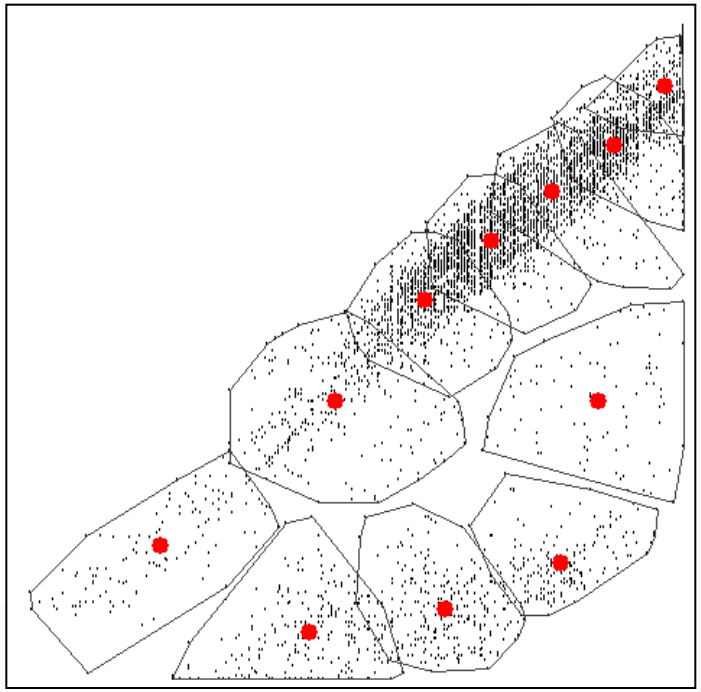
Color quantization of images

Color image

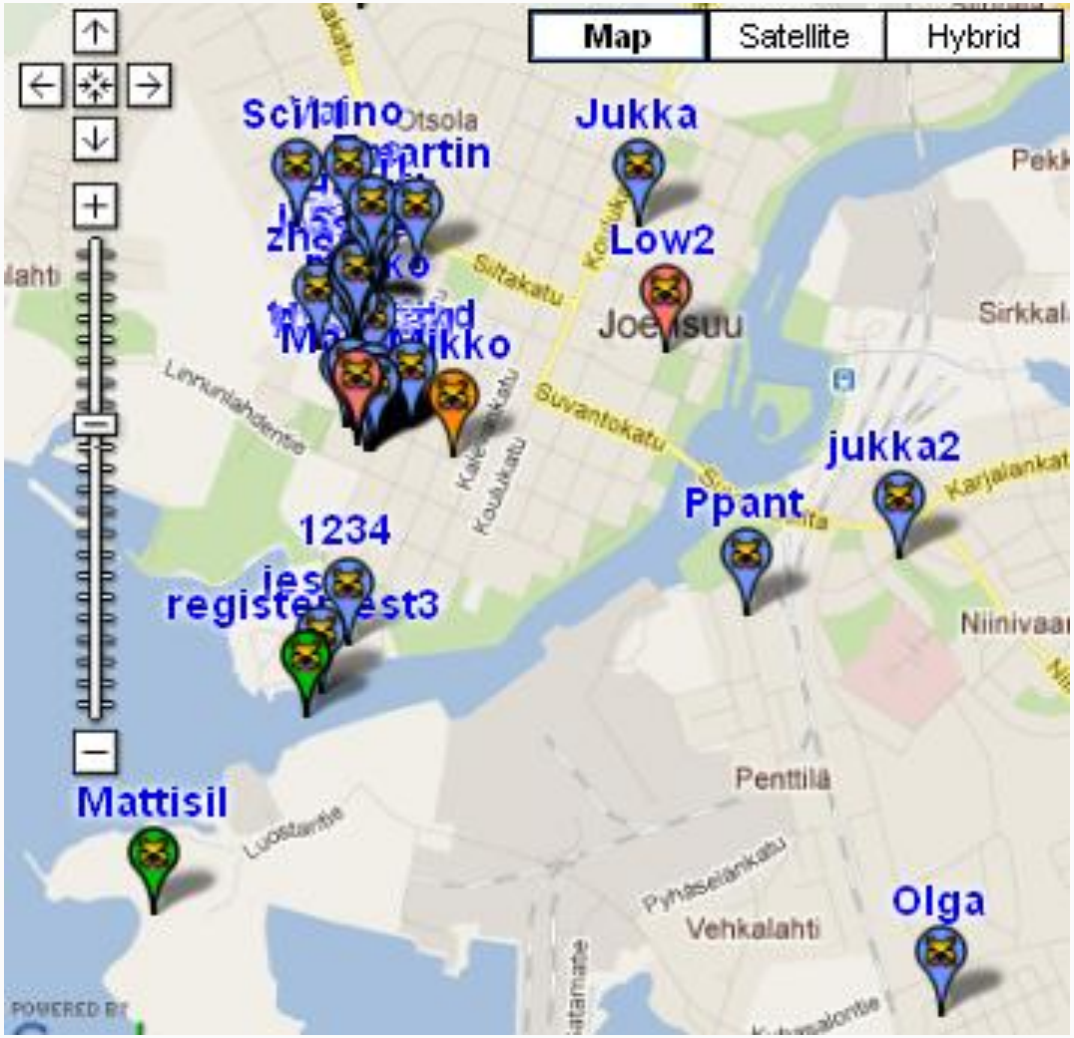
RGB samples

```
VQ TRAINING SET 2.0
5400 vectors, total frequency 5400, vector size 3
Values: 0...255 (1 byte per element)
Generated by TXT2CB
138 149 107  (1)
113 125  84  (1)
110 125  88  (1)
111 130  95  (1)
 94 119  84  (1)
 96 125  89  (1)
105 138  97  (1)
 96 133  84  (1)
 93 131  76  (1)
 99 135  75  (1)
 95 128  67  (1)
 88 117  57  (1)
101 123  68  (1)
126 142  92  (1)
134 147 102  (1)
128 135  92  (1)
171 167 125  (1)
178 168 126  (1)
188 178 136  (1)
198 188 146  (1)
202 190 151  (1)
201 189 150  (1)
205 193 154  (1)
213 200 161  (1)
223 212 174  (1)
213 205 166  (1)
216 208 169  (1)
215 211 171  (1)
208 204 166  (1)
211 208 170  (1)
```

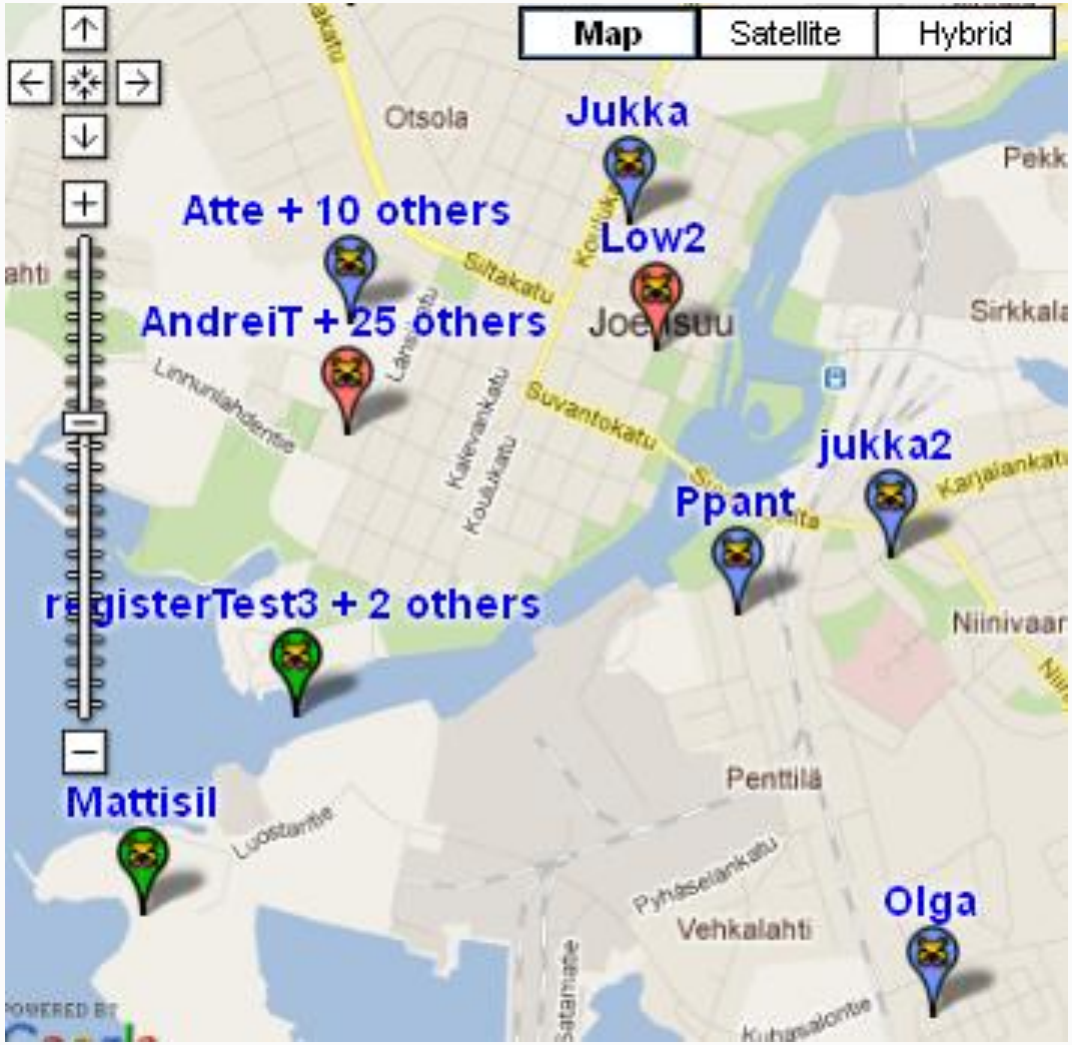
Clustering



Users on Map


Map
Satellite
Hybrid
Jukka
Low2
Joensuu
jukka2
Ppant
1234
Mattisil
Olga
Siitakatu
Suvantokatu
Linnunlahdentie
Koulukatu
Penttilä
Vehkalahti
Pyhäseläntie
Luostarintie
Niinivaara
Sirkkala
Karjalankatu

Clustering the Users



Clustering of Photos in Two Ways

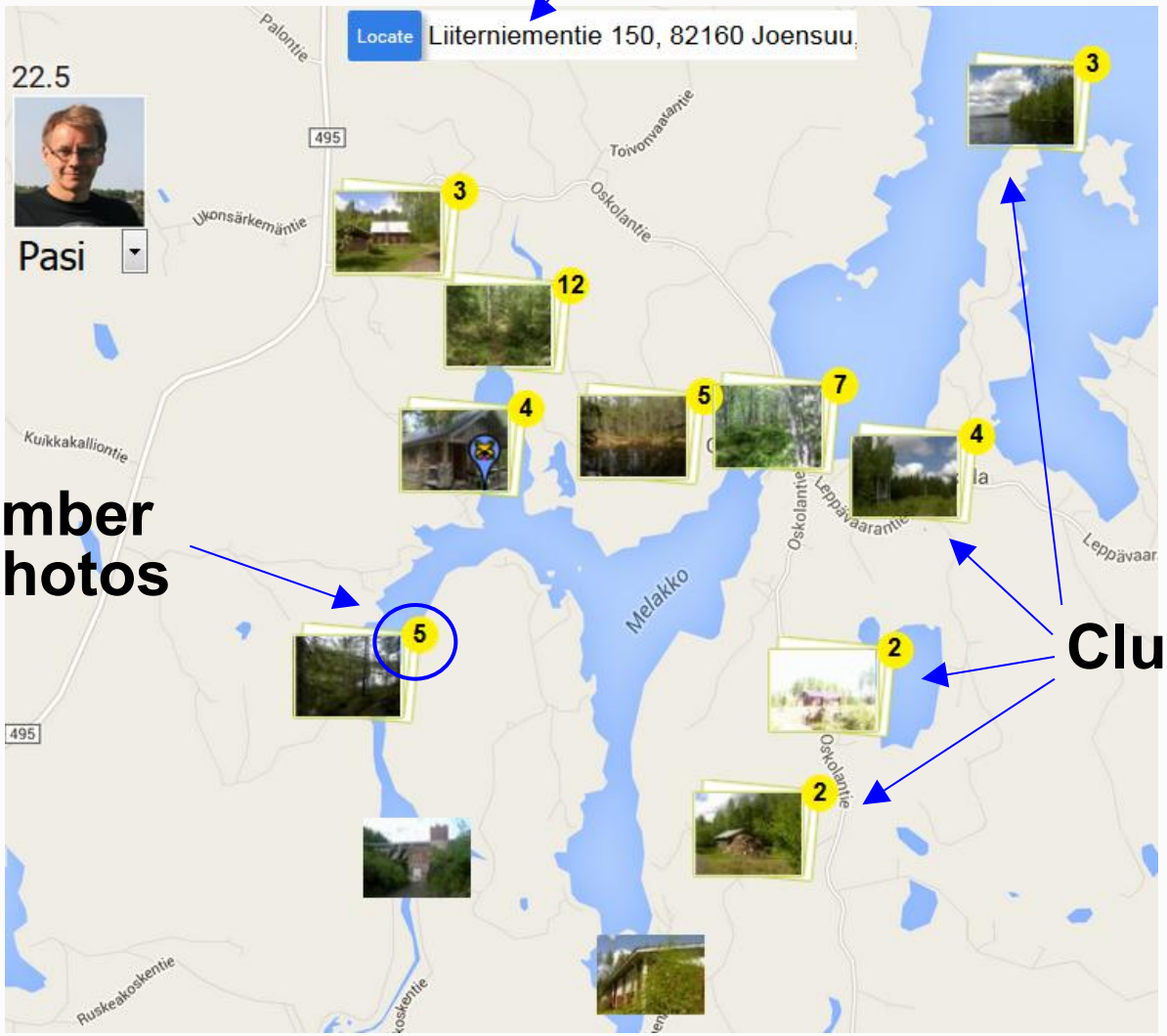
Clustering timeline

Clustering of photos

Photo Clusters on Map

User and date

Last known location of the user

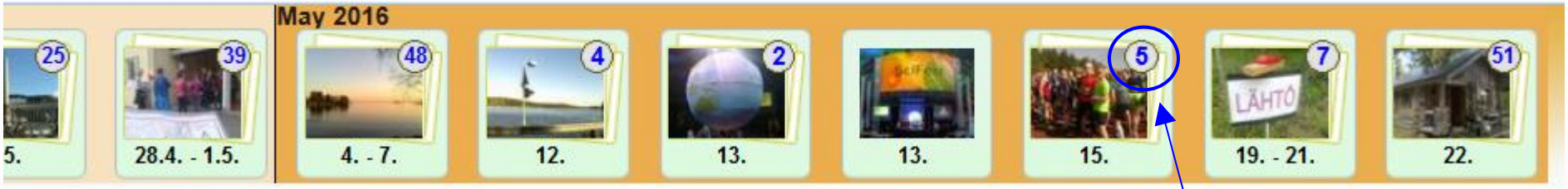


Number of photos

Clusters

Locate
Kerkonkoskentie 1121, 77700 Rauta
Map
Satellite
OSM
KR
1.1.2000 - 13.3.2015
Pasi
9938
Juuka
Siilinjärvi
Juankoski
Kallavesi
Kaavi
Riistavesi
Kotasaari
Hiltulanlahti
Vehmersalmi
Polvijärvi
Höytiäinen
Viinijärvi
Ylämylly
Joensuu
Reijola
Pyhäselkä
Kermajärvi
MOPSI v0.62 12.2.2015

Clusters in the timeline view



Clusters

Number of photos

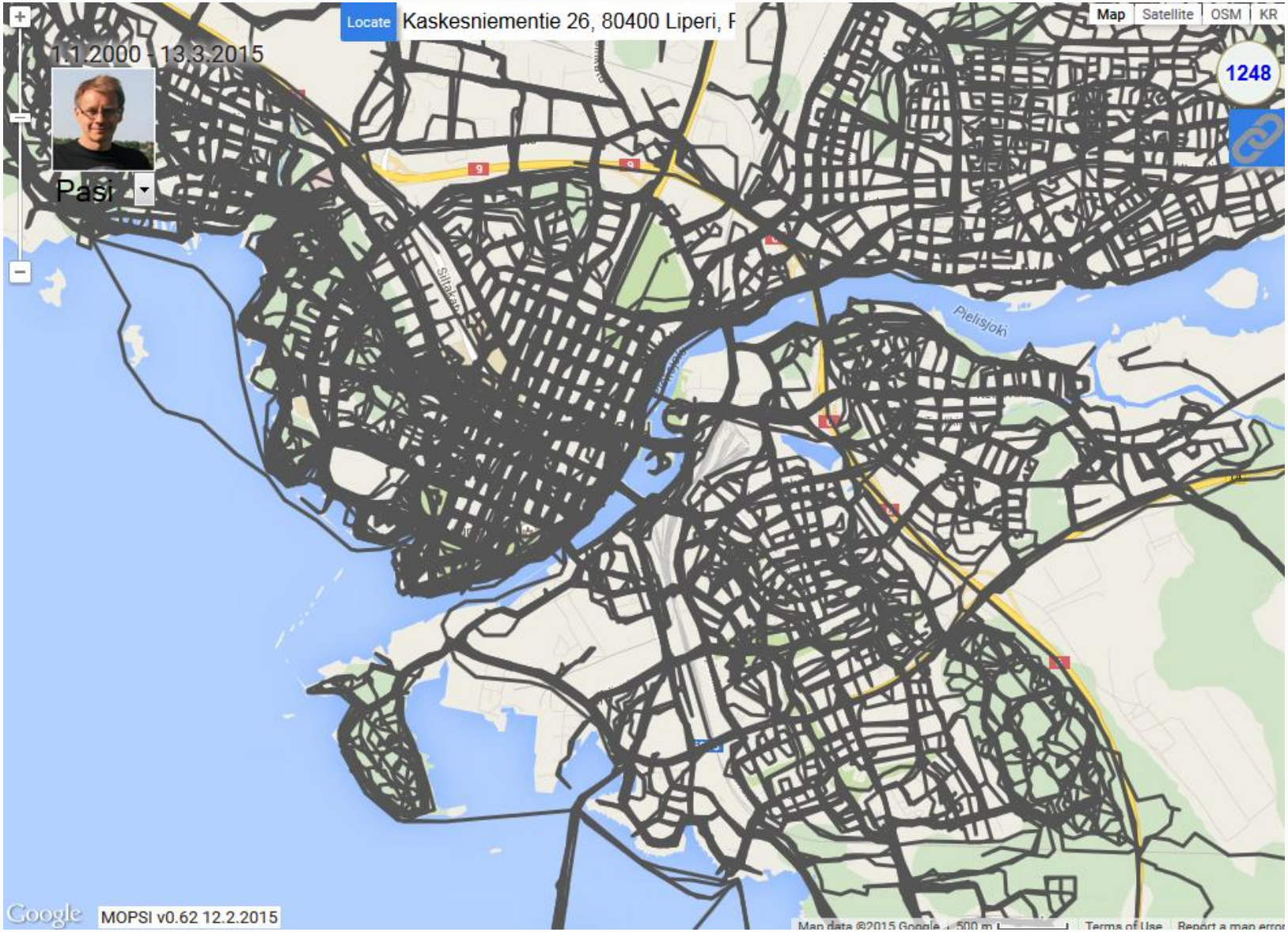
Functions:

Open cluster

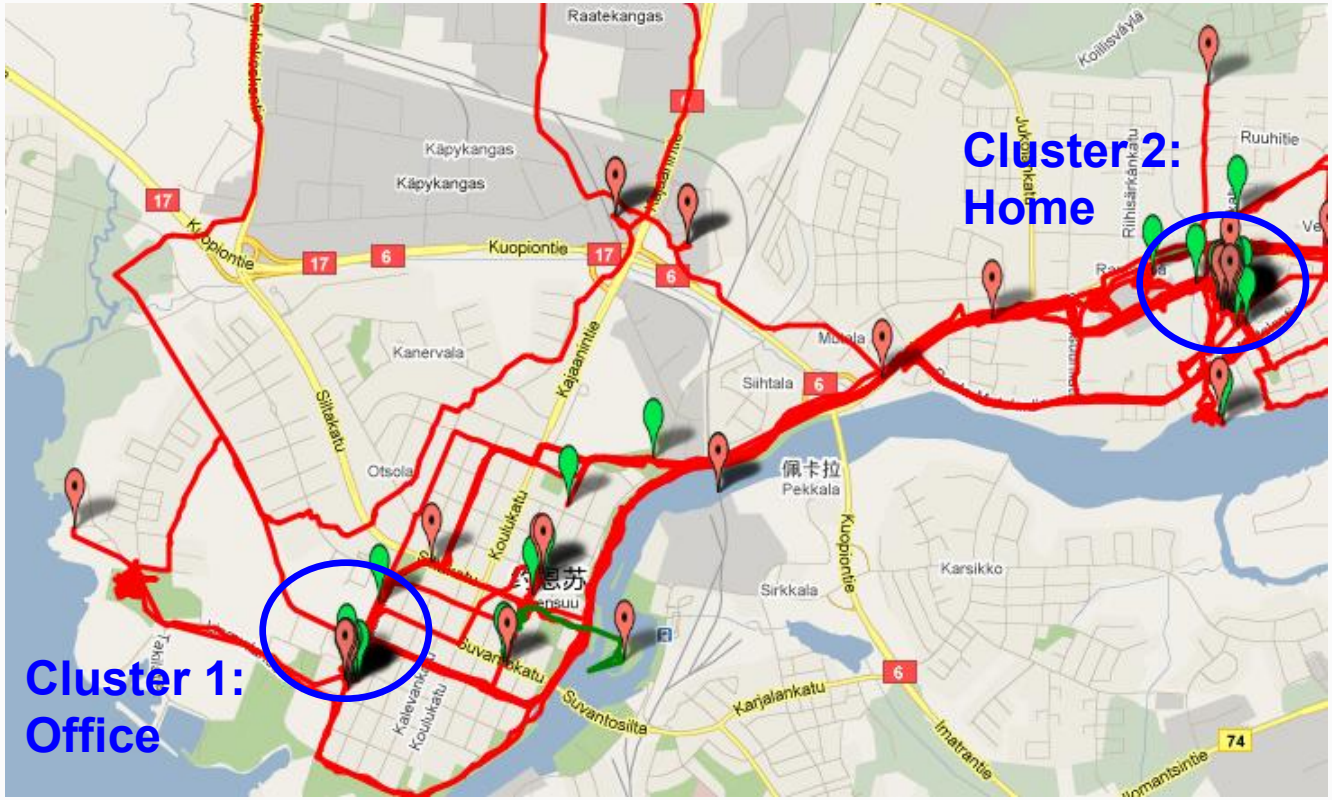
Start slideshow

Clustering GPS tracks

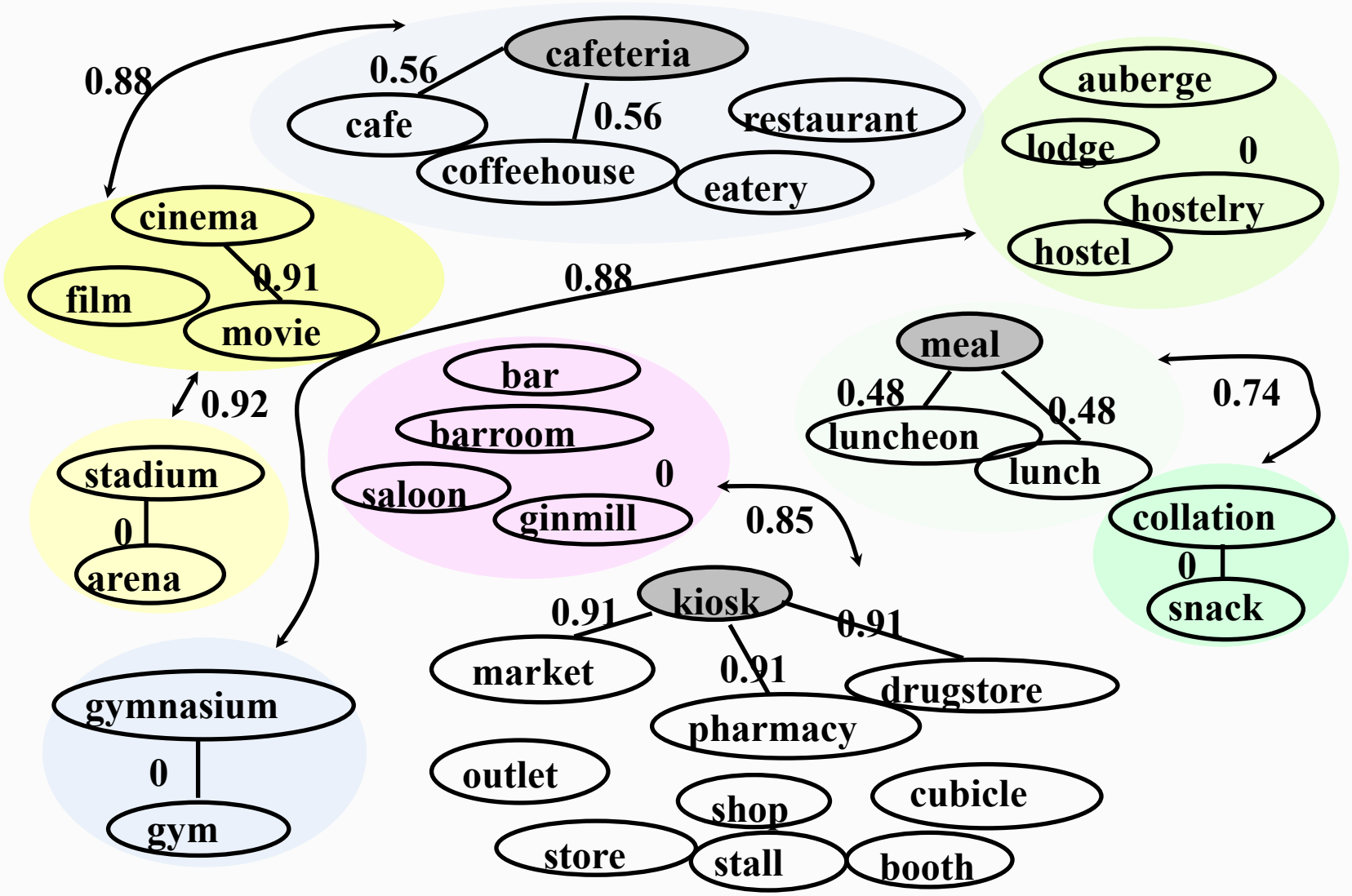
Mobile users, taxi routes, fleet management



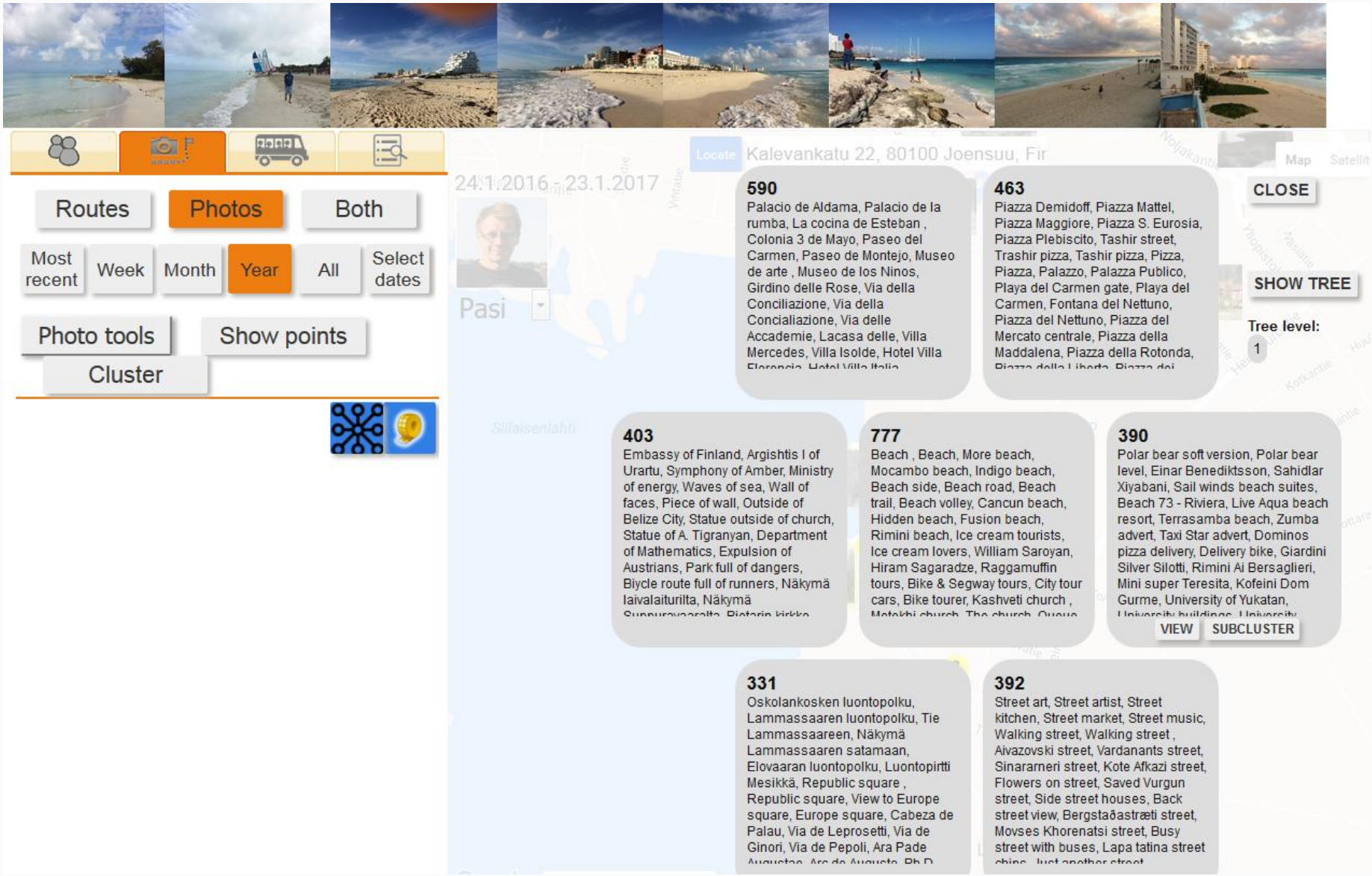
Conclusions from Clusters



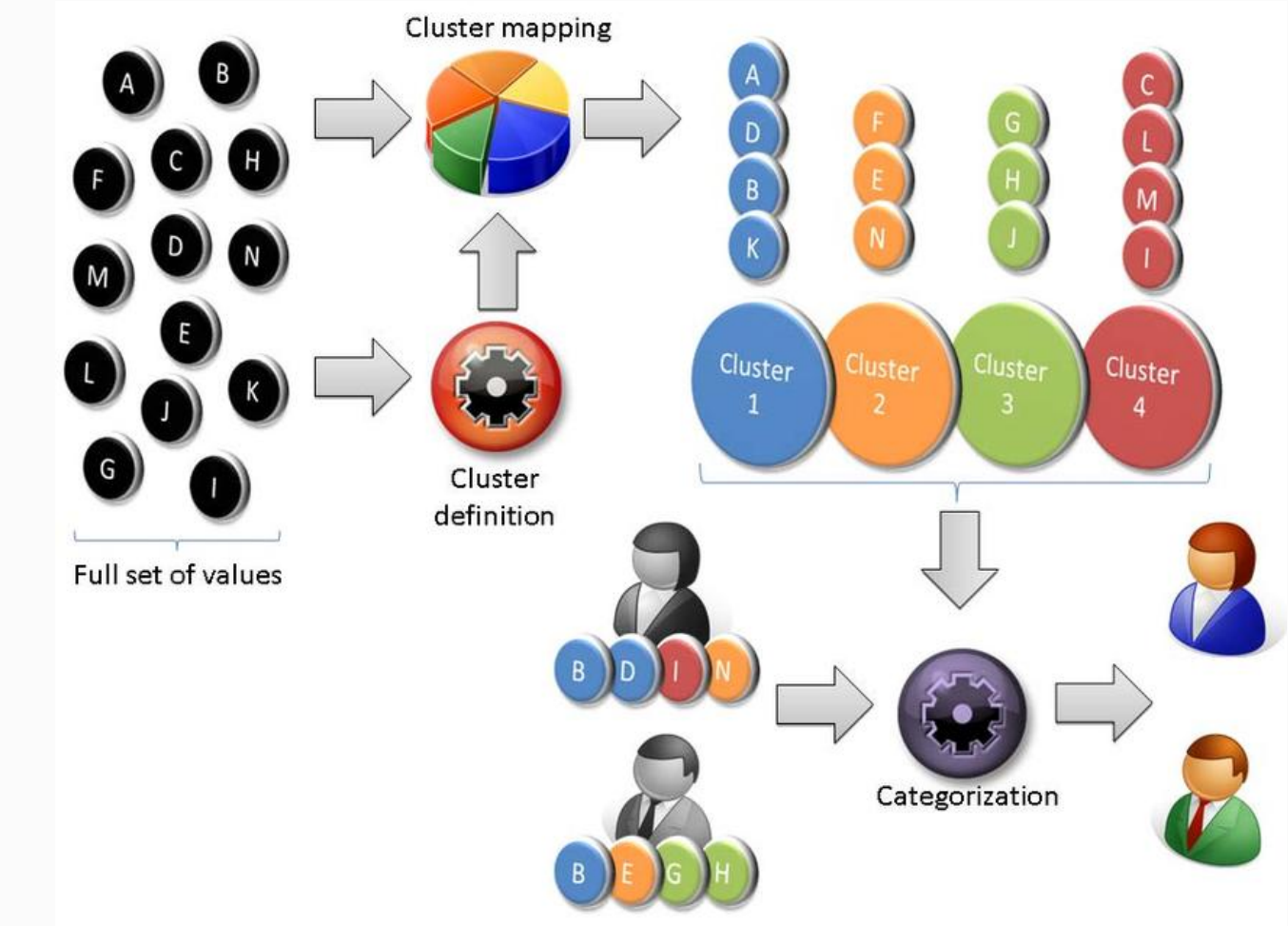
Clustering keywords



Clustering Text Descriptions



Clustering User Preferences



Clustering problem

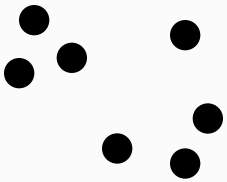
Subproblems of clustering

1. Where are the clusters?
   (Algorithmic problem)
2. How many clusters?
   (Methodological problem: which criterion?)
3. Selection of attributes
   (Application related problem)
4. Preprocessing the data
   (Practical problems: normalization, outliers)

Definitions and data

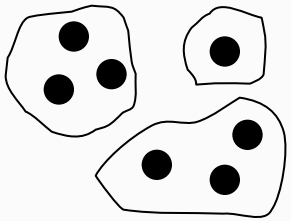
Set of *N* data points:

$X=\{x_1, x_2, \ldots, x_N\}$



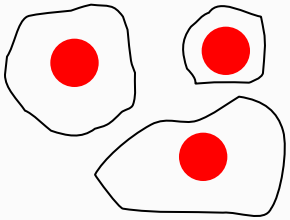
Partition of the data:

$P=\{p_1, p_2, \ldots, p_M\}$,



Set of *M* cluster prototypes (centroids):

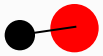
$C=\{c_1, c_2, \ldots, c_M\}$,



Distance and cost function

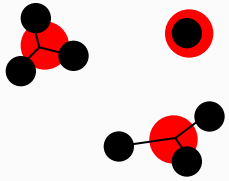
Euclidean distance of data vectors:

$$d(x_i, x_j) = \sqrt{\sum_{k=1}^{K} \left(x_i^k - x_j^k \right)^2}$$



Total square error:

$$TSE(C, P) = \sum_{i=1}^{N} \left\| x_i - c_{p_i} \right\|^2$$



K-means algorithm

K-means algorithm

X = Data set
C = Cluster centroids
P = Partition

K-Means(X, C) $\rightarrow$ (C, P)

REPEAT

$C_{prev} \leftarrow C$;

FOR all $i \in [1, N]$ DO
$p_i \leftarrow$ FindNearest(x_i, C);

Optimal partition

FOR all $j \in [1, k]$ DO
$c_j \leftarrow$ Average of $x_i \ \forall \ p_i = j$;

Optimal centoids

UNTIL $C = C_{prev}$

Summary

How to solve?

Solve the clustering:

- Given input data (X) of N data vectors, and number of clusters (M), find the clusters.
- Result given as a set of prototypes, or partition.

Algorithmic problem

Solve the number of clusters:

- Define appropriate cluster validity function f.
- Repeat the clustering algorithm for several M.
- Select the best result according to f.

Mathematical problem

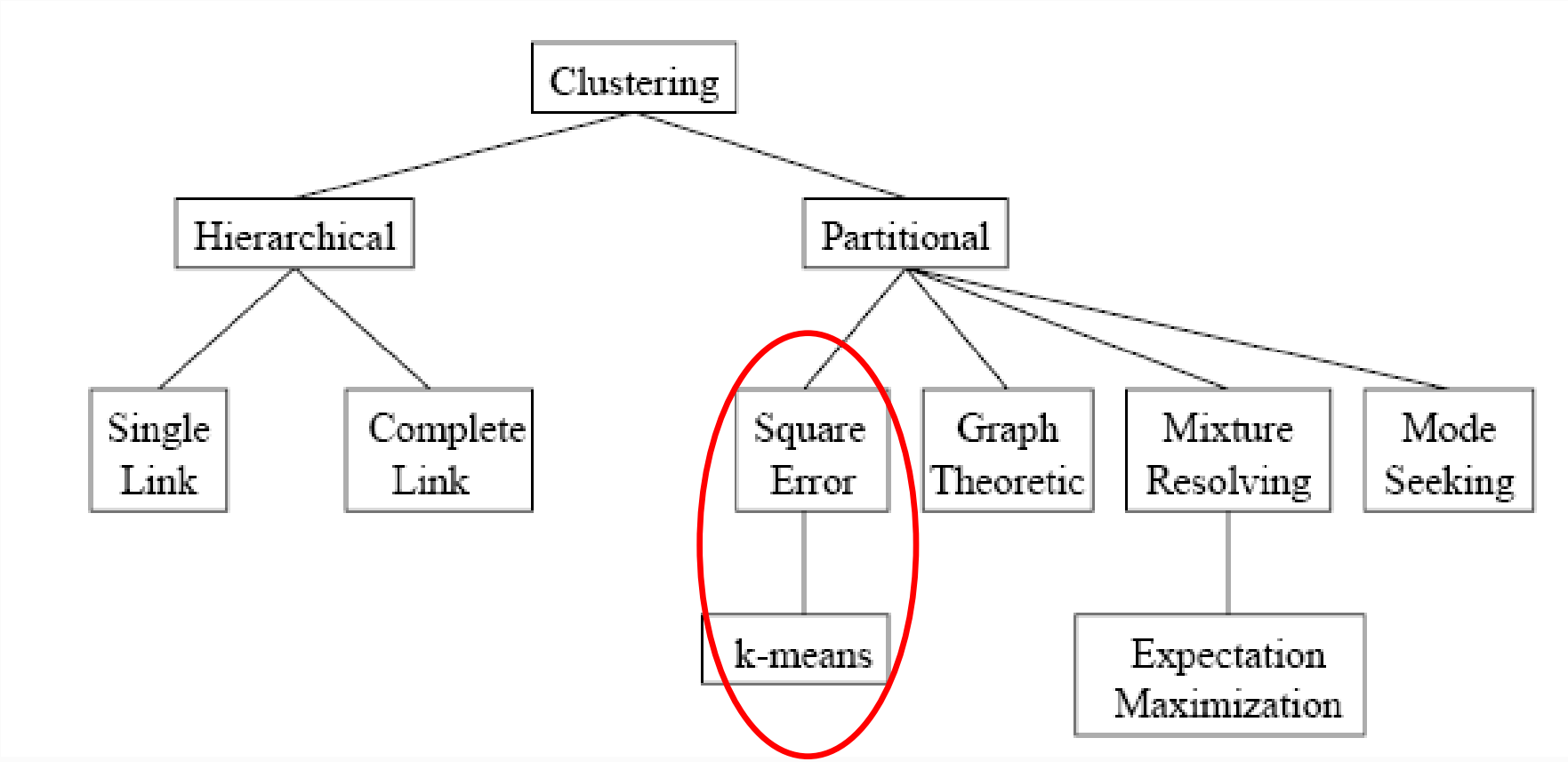
Solve the problem efficiently.

Computer science problem

Taxonomy of clustering

[Jain, Murty, Flynn, Data clustering: A review, *ACM Computing Surveys*, 1999.]

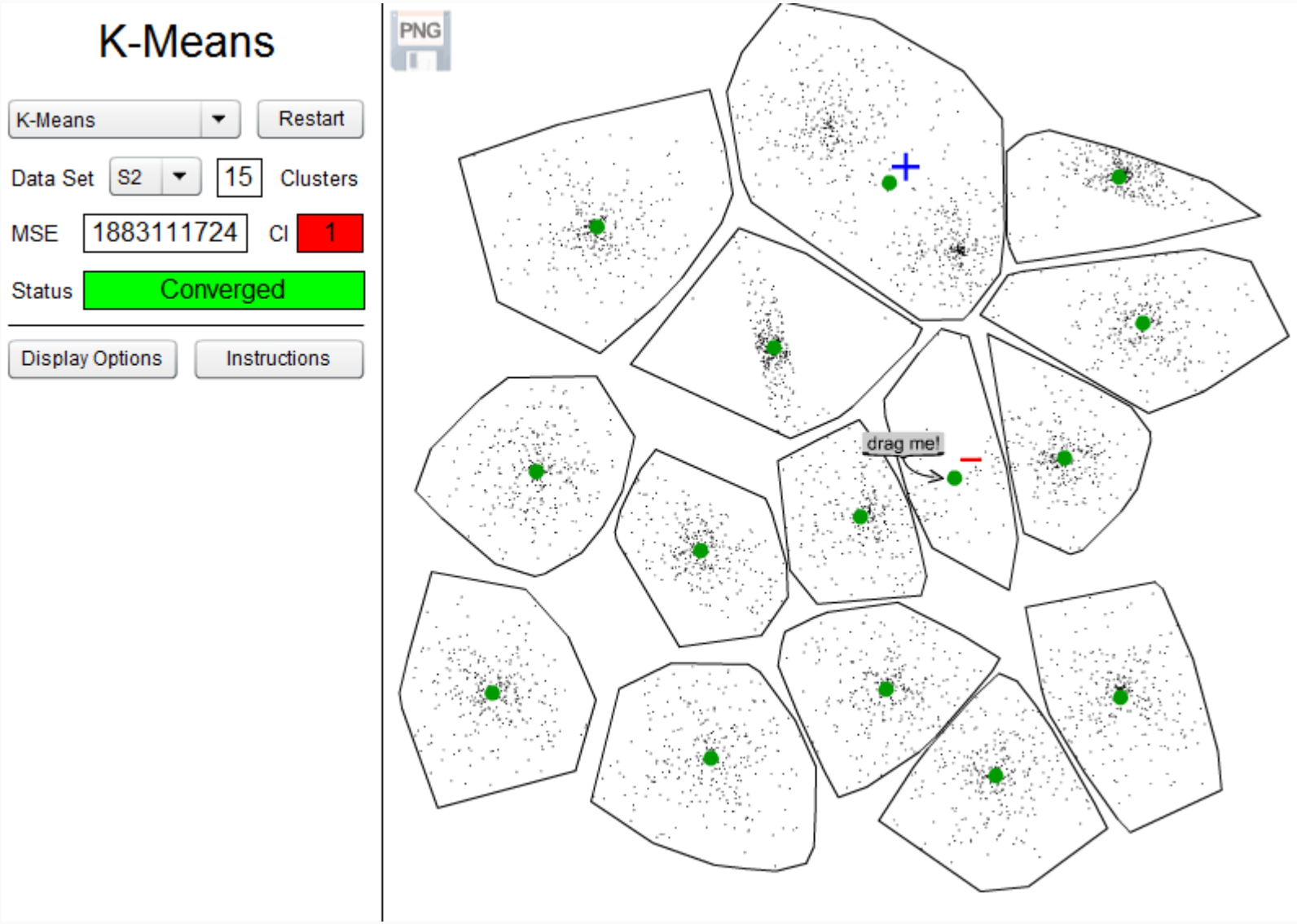
- One possible classification based on cost function.
- MSE is well defined and most popular.



Software

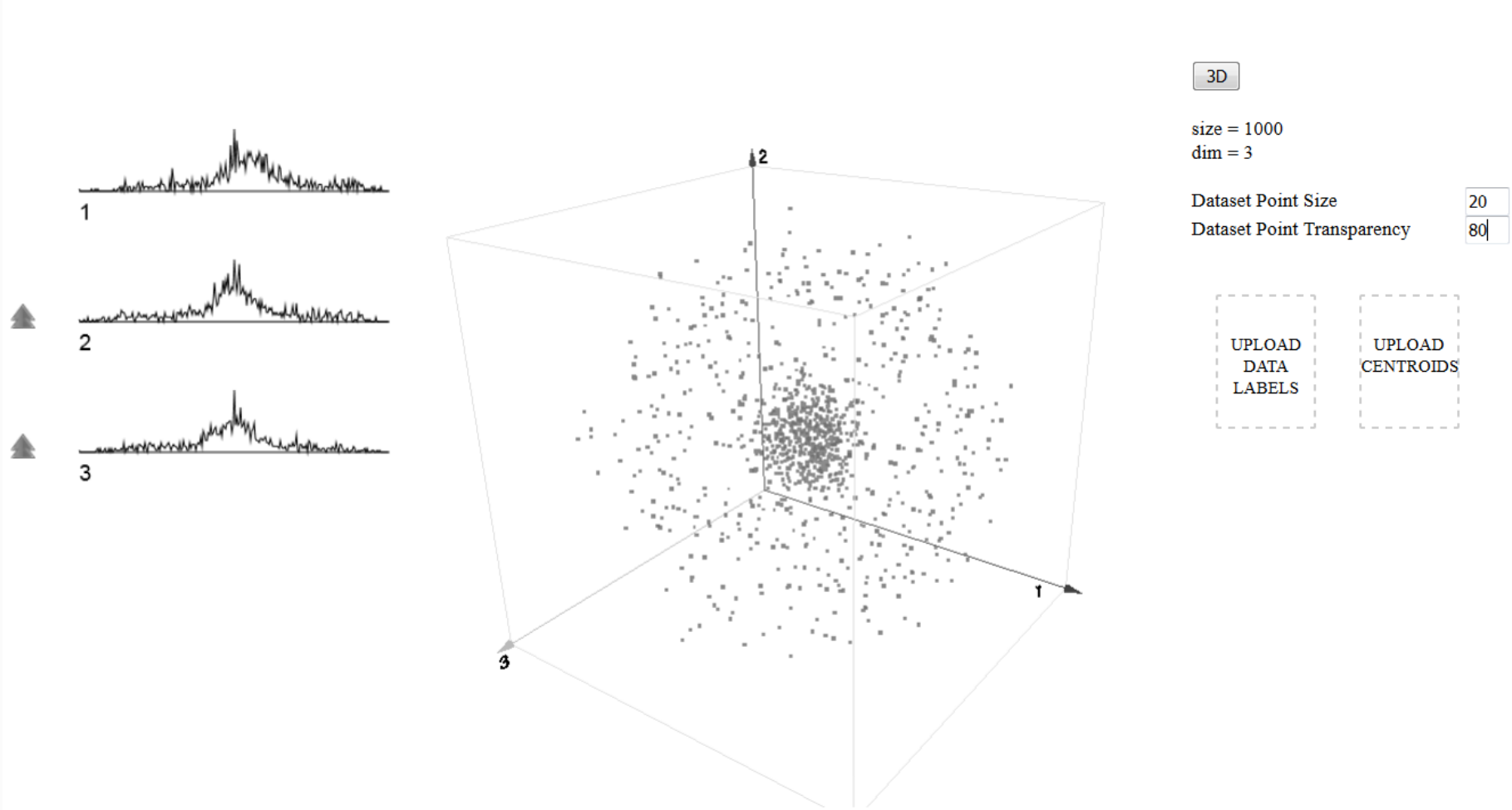
Animator

http://cs.uef.fi/sipu/clustering/animator/



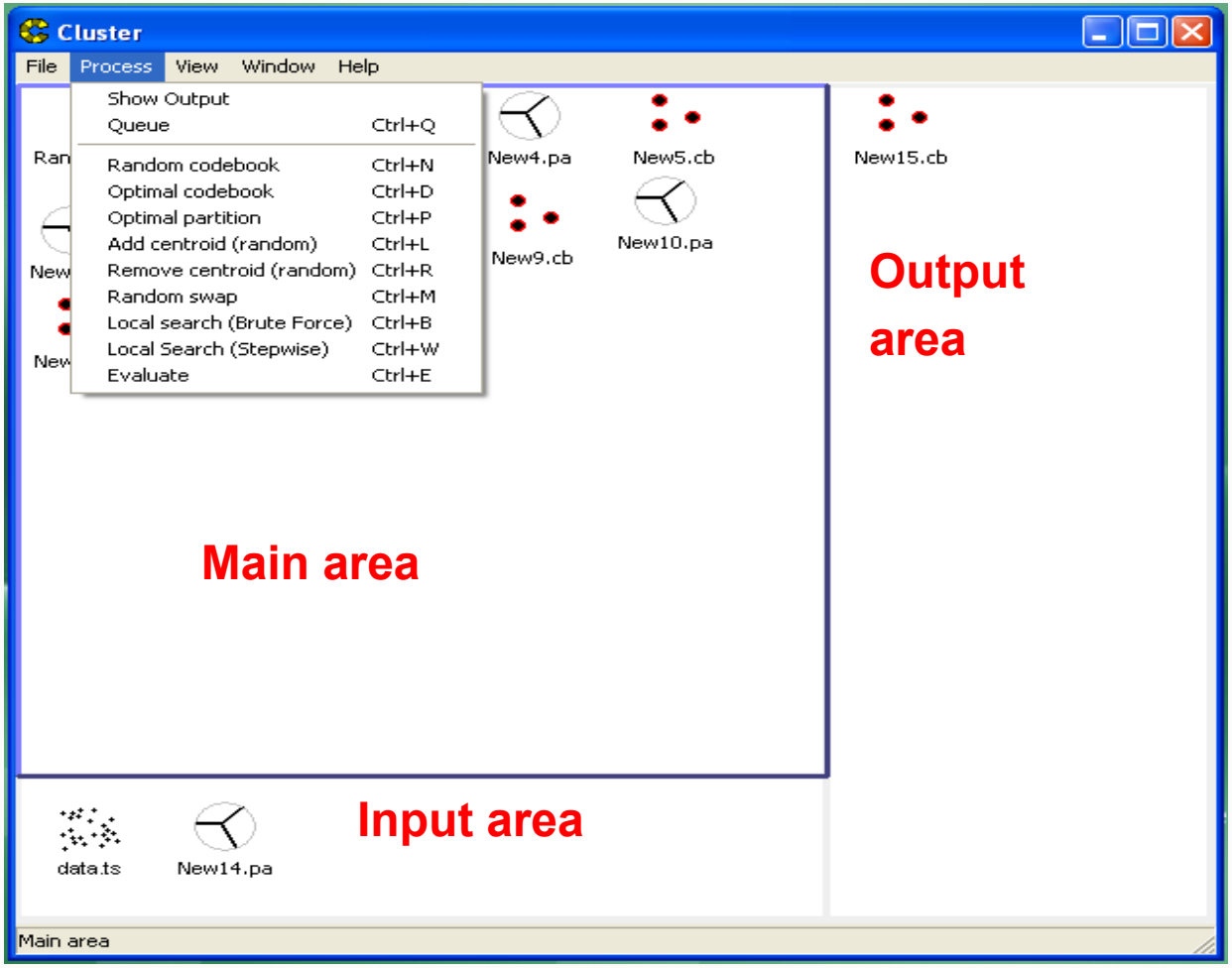
Clusterator

http://cs.uef.fi/paikka/Radu/clusterator/



Cluster software

http://cs.uef.fi/sipu/soft/cluster2009.exe



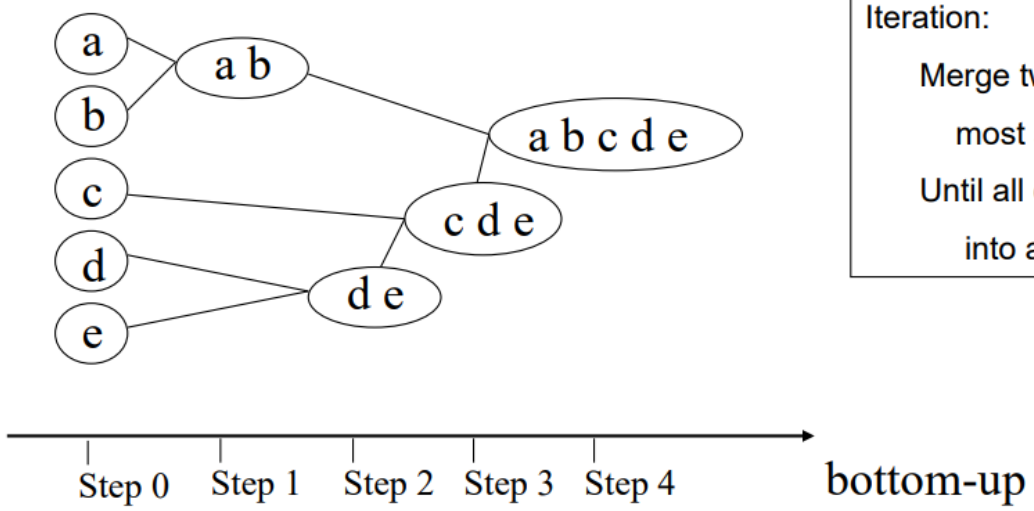
- **Main area**: working space for data
- **Input area**: inputs to be processed
- **Output area**: obtained results
- **Menu Process**: selection of operation

Literature

1. S. Theodoridis and K. Koutroumbas, Pattern Recognition, Academic Press, 3rd edition, 2006.
2. C. Bishop, Pattern Recognition and Machine Learning, Springer, 2006.
3. A.K. Jain, M.N. Murty and P.J. Flynn, Data clustering: A review, *ACM Computing Surveys*, 31(3): 264-323, September 1999.
4. M.R. Garey, D.S. Johnson and H.S. Witsenhausen, The complexity of the generalized Lloyd-Max problem, *IEEE Transactions on Information Theory*, 28(2): 255-256, March 1982.
5. F. Aurenhammer: Voronoi diagrams-a survey of a fundamental geometric data structure, *ACM Computing Surveys*, 23 (3), 345-405, September 1991.

Hierarchical Clustering

- **Agglomerative approach**



Initialization:

Each object is a cluster

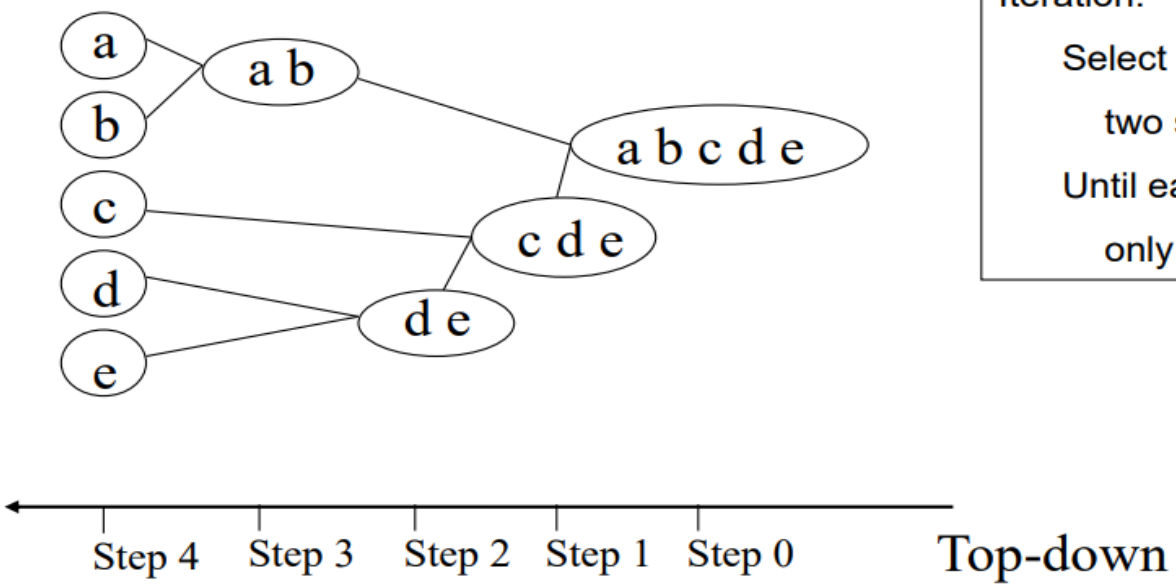
Iteration:

Merge two clusters which are most similar to each other;

Until all objects are merged into a single cluster

Hierarchical Clustering

- **Divisive Approaches**



Initialization:

All objects stay in one cluster

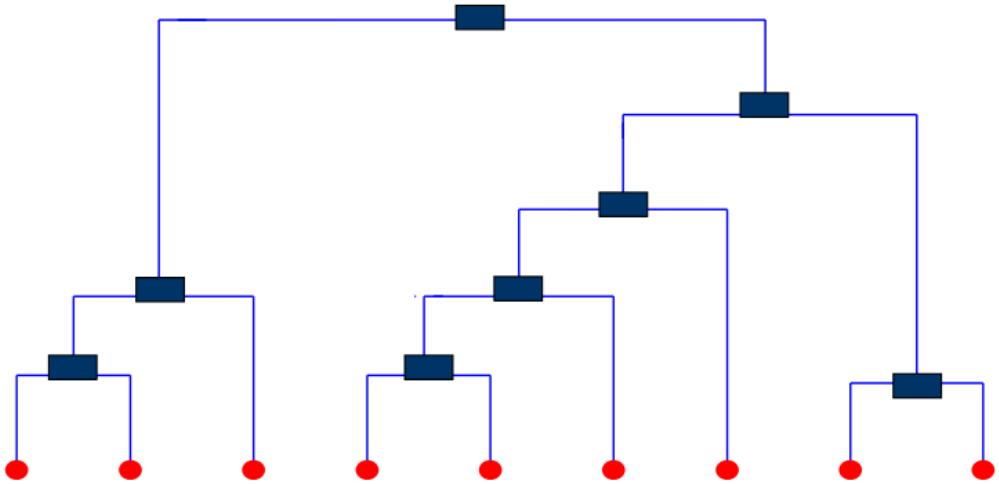
Iteration:

Select a cluster and split it into two sub clusters

Until each leaf cluster contains only one object

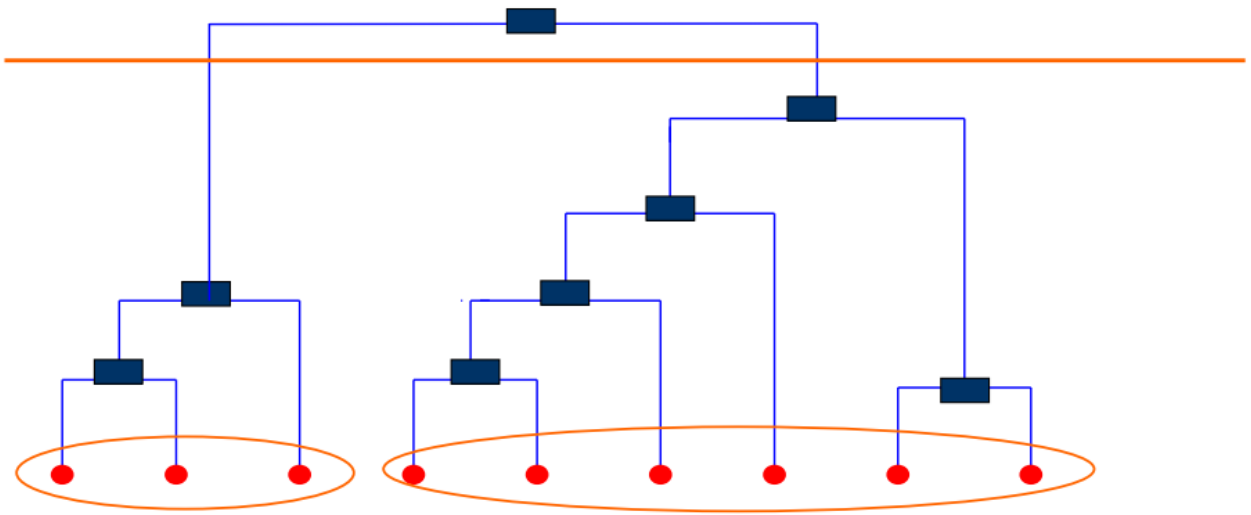
Dendrogram

- A tree that shows how clusters are merged/split hierarchically
- Each node on the tree is a cluster; each leaf node is a singleton cluster



Dendrogram

- A clustering of the data objects is obtained by cutting the *dendrogram* at the desired level, then each connected component forms a cluster

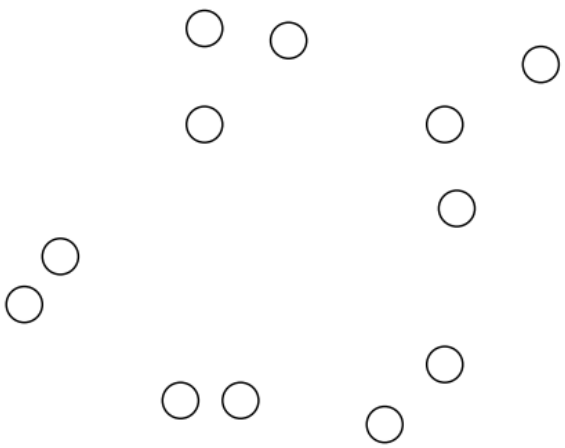


Agglomerative Clustering Algorithm

- More popular hierarchical clustering technique
- Basic algorithm is straightforward
  1. Compute the distance matrix
  2. Let each data point be a cluster
  3. **Repeat**
  4. Merge the two closest clusters
  5. Update the distance matrix
  6. **Until** only a single cluster remains
- Key operation is the computation of the distance between two clusters
  - Different approaches to defining the distance between clusters distinguish the different algorithms

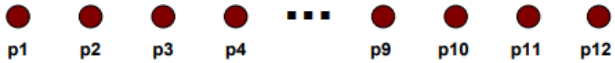
Starting Situation

- Start with clusters of individual points and a distance matrix



	p1	p2	p3	p4	p5	. . .
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Distance Matrix

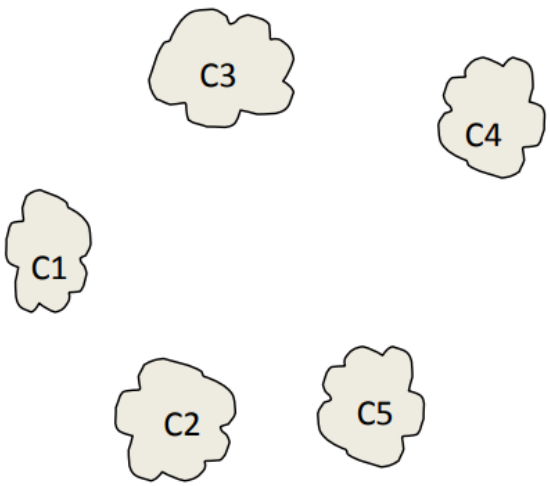


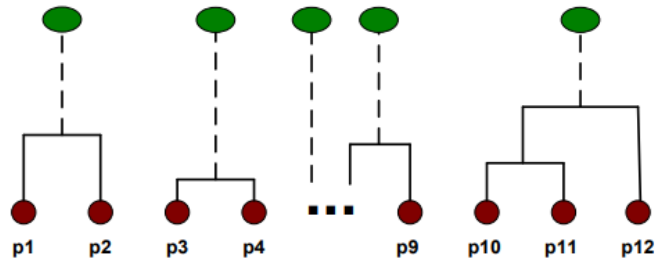
Intermediate Situation

- After some merging steps, we have some clusters
- Choose two clusters that has the smallest distance (largest similarity) to merge

	C1	C2	C3	C4	C5
C1					
C2					
C3					
C4					
C5					

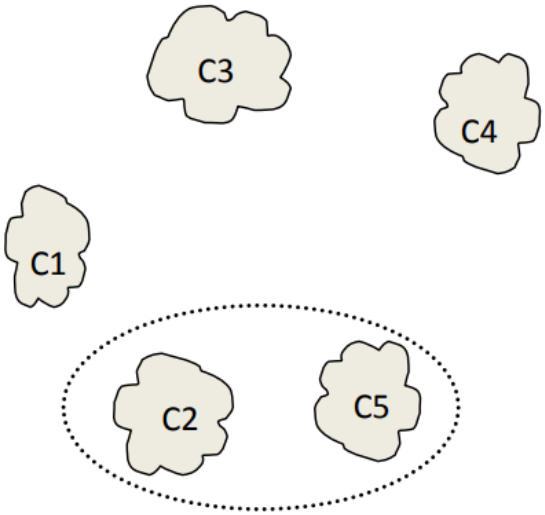
Distance Matrix

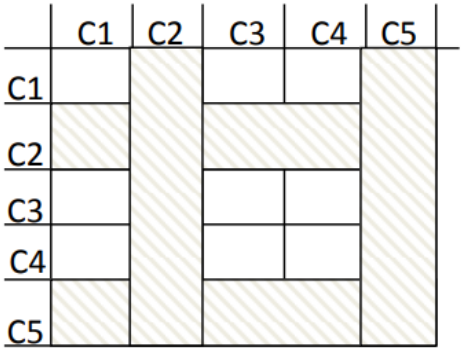




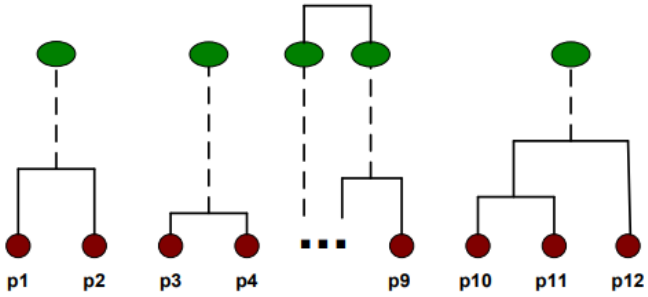
Intermediate Situation

- We want to merge the two closest clusters (C2 and C5) and update the distance matrix.



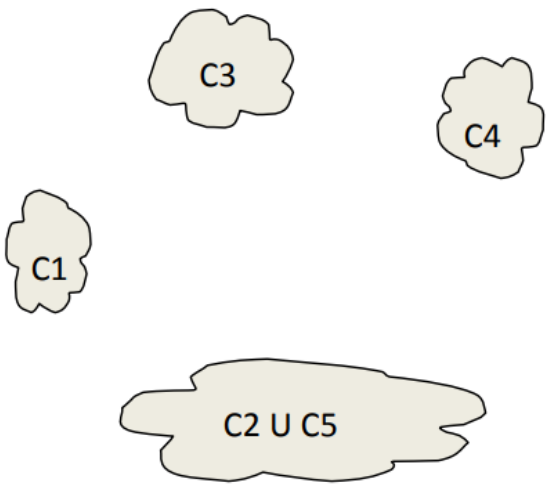


Distance Matrix



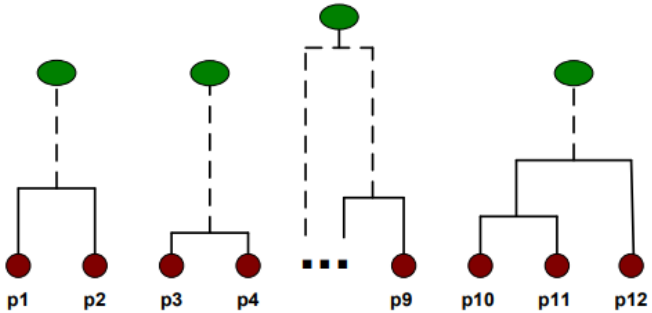
After Merging

- The question is "How do we update the distance matrix?"

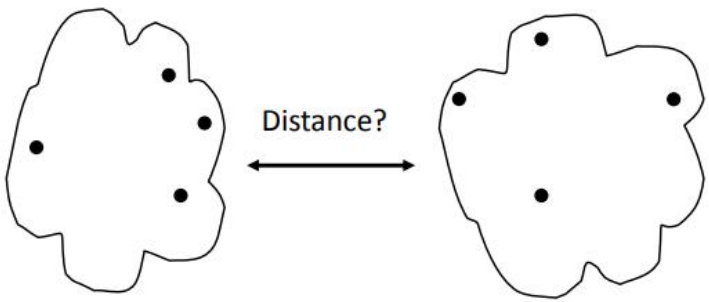


	C1	C2 U C5	C3	C4
C1		?		
C2 U C5	?	?	?	?
C3		?		
C4		?		

Distance Matrix



How to Define Inter-Cluster Distance



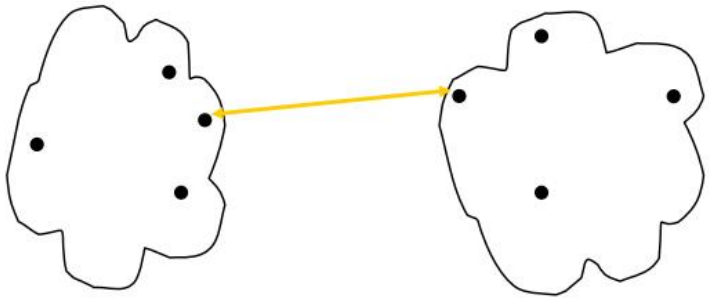
	p1	p2	p3	p4	p5	. . .
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Distance Matrix

- MIN
- MAX
- Group Average
- Distance Between Centroids
- ……

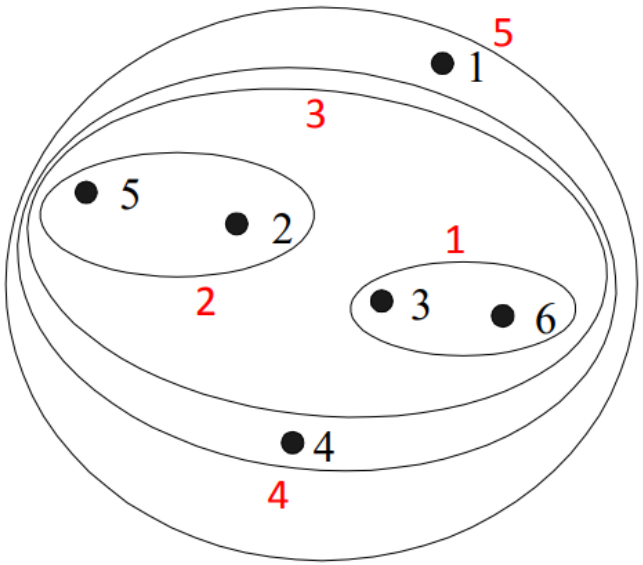
MIN or Single Link

- **Inter-cluster distance**
  - The distance between two clusters is represented by the distance of the *closest pair of data objects* belonging to different clusters.
  - Determined by one pair of points, i.e., by one link in the proximity graph

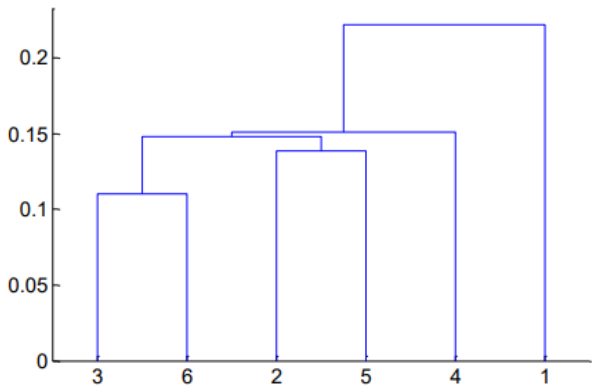


$$d_{\min}(C_i, C_j) = \min_{p \in C_i, q \in C_j} d(p, q)$$

MIN

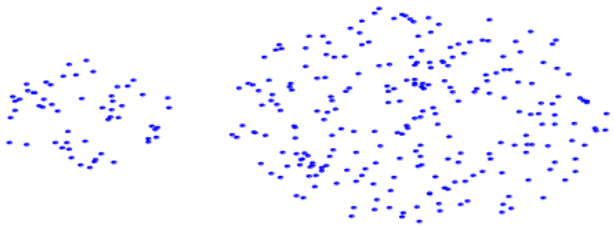


Nested Clusters

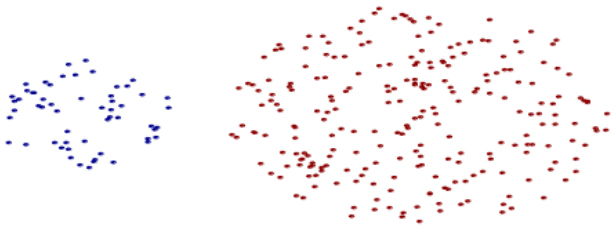


Dendrogram

Strength of MIN



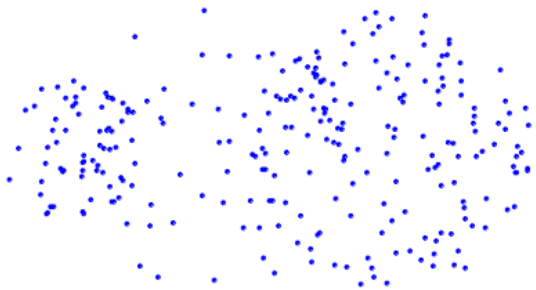
Original Points



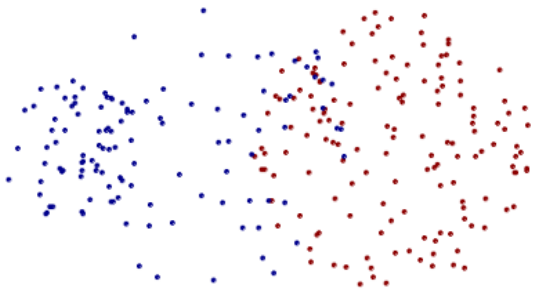
Two Clusters

- Can handle non-elliptical shapes

Limitations of MIN



Original Points

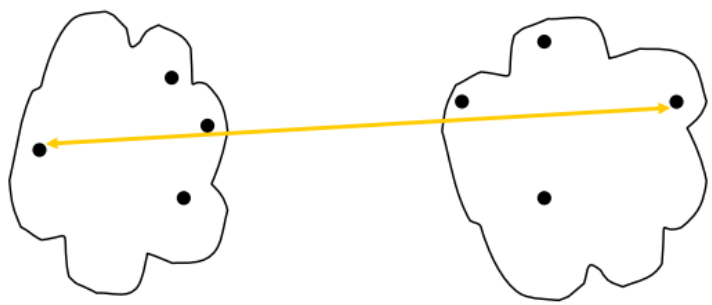


Two Clusters

- Sensitive to noise and outliers

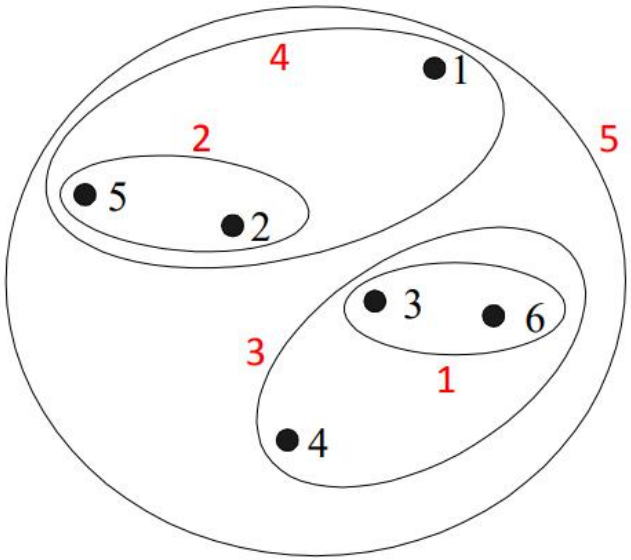
MAX or Complete Link

- **Inter-cluster distance**
  - The distance between two clusters is represented by the distance of the *farthest pair of data objects* belonging to different clusters

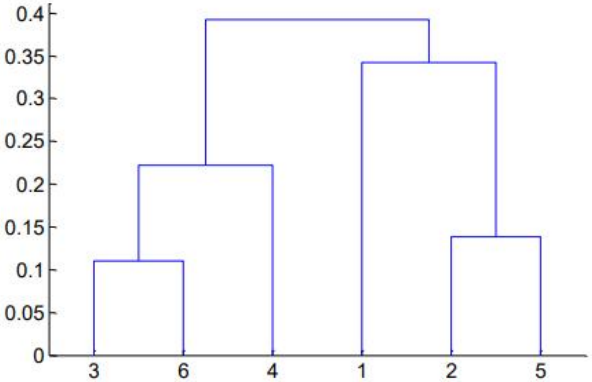


$$d_{\min}(C_i, C_j) = \max_{p \in C_i, q \in C_j} d(p, q)$$

MAX

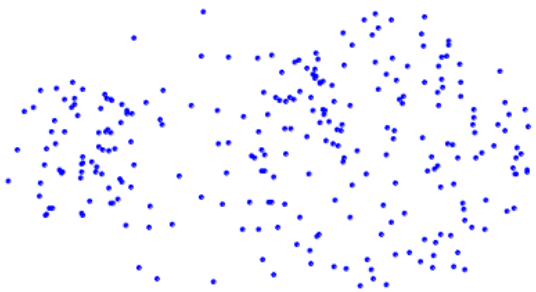


Nested Clusters

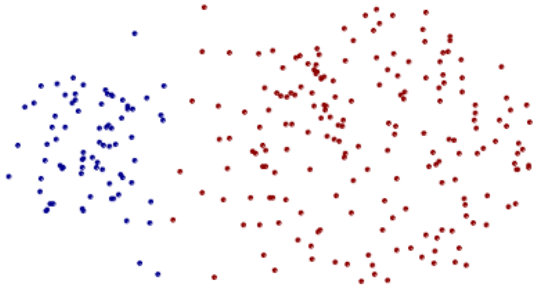


Dendrogram

Strength of MAX



Original Points

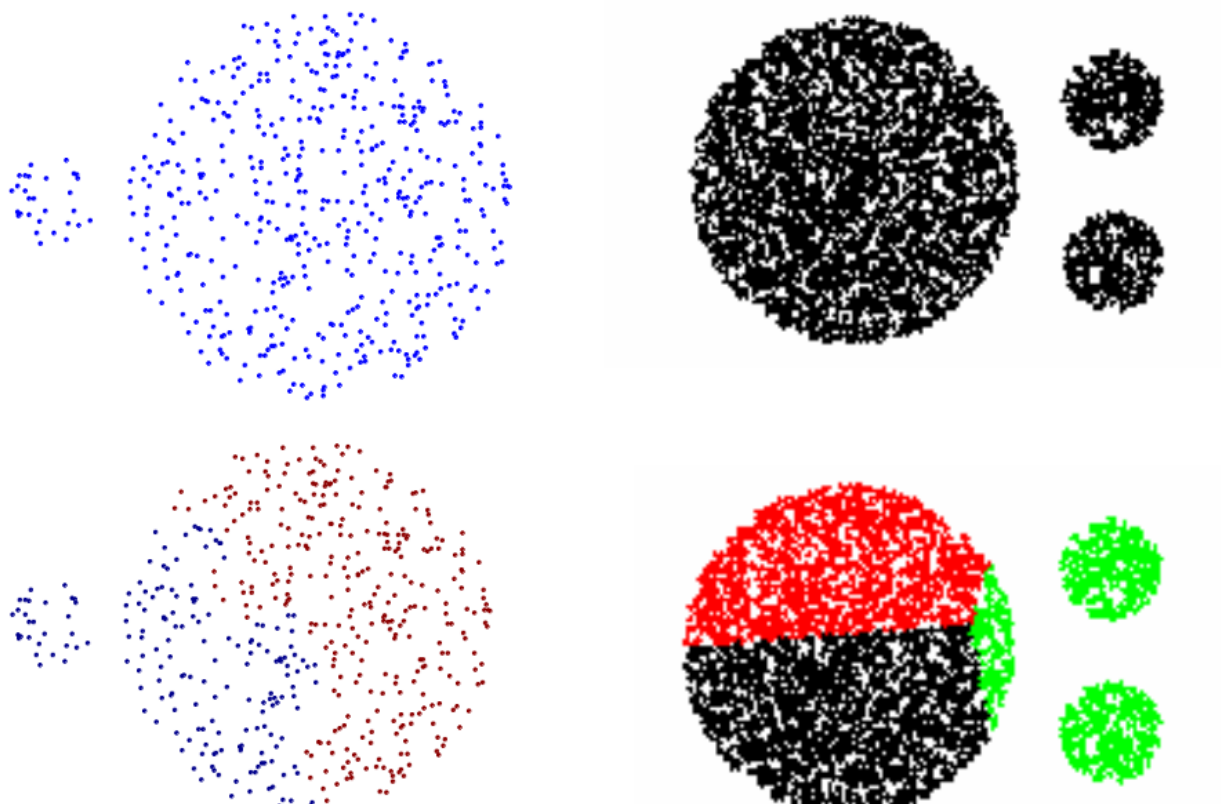


Two Clusters

- Less susceptible to noise and outliers

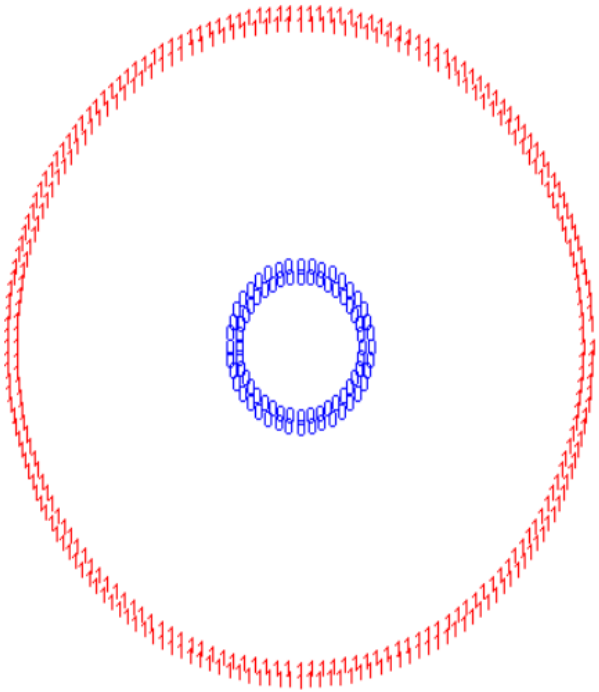
Limitations of MAX

- Tends to break large clusters

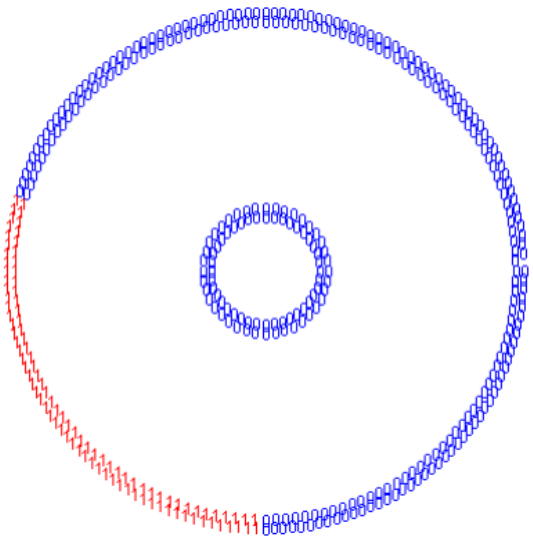


The MAX approach can cause "over-splitting" of large clusters, especially if they have outliers or loosely connected regions. The clustering might separate them incorrectly, as shown in the third part of the image where a red and green section are formed within the original large cluster.

Limitations of MAX



MIN (2 clusters)

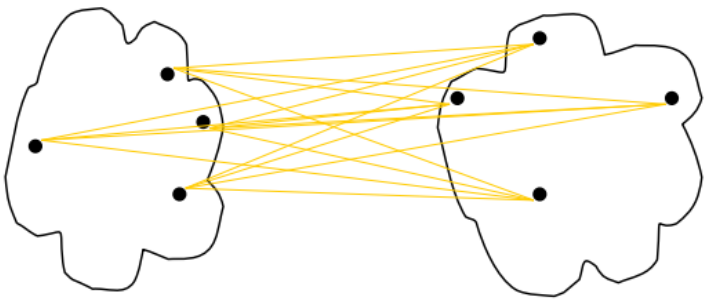


MAX (2 clusters)

- Biased towards globular clusters

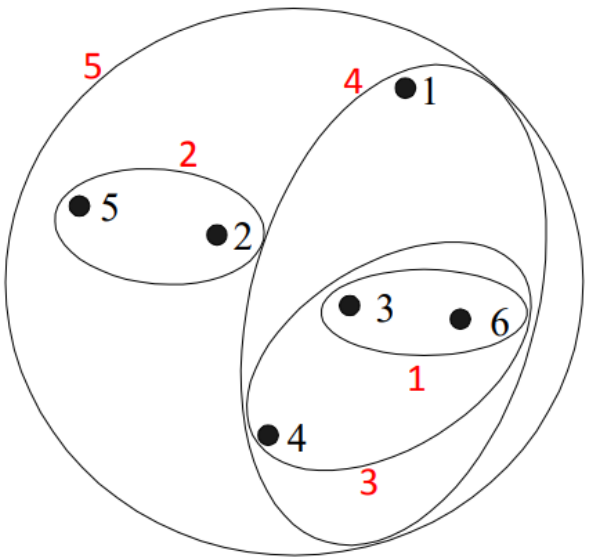
Group Average or Average Link

- **Inter-cluster distance**
  - The distance between two clusters is represented by the *average* distance of *all pairs of data objects* belonging to different clusters
  - Determined by all pairs of points in the two clusters

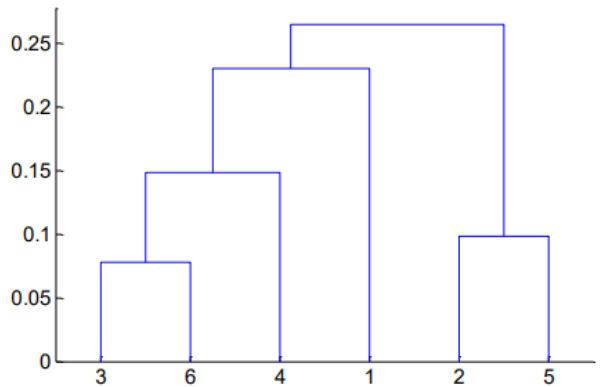


$$d_{\min}(C_i, C_j) = \underset{p \in C_i, q \in C_j}{avg} \; d(p,q)$$

Group Average



Nested Clusters



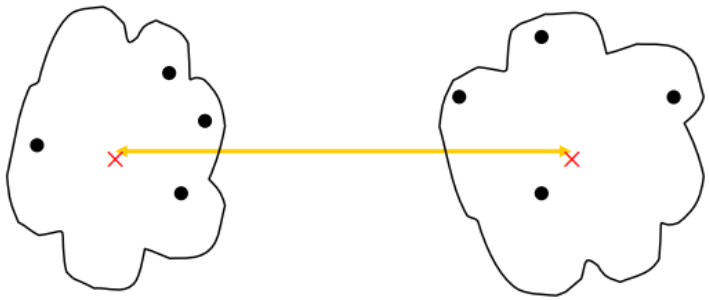
Dendrogram

Group Average

- Compromise between Single and Complete Link
- Strengths
  - Less susceptible to noise and outliers
- Limitations
  - Biased towards globular clusters

Centroid Distance

- **Inter-cluster distance**
  - The distance between two clusters is represented by the distance between *the centers of the clusters*
  - Determined by cluster centroids



$$d_{mean}(C_i, C_j) = d(m_i, m_j)$$

Ward's Method

- Similarity of two clusters is based on the increase in squared error when two clusters are merged
  - Similar to group average if distance between points is distance squared
- Less susceptible to noise and outliers
- Biased towards globular clusters
- Hierarchical analogue of K-means
  - Can be used to initialize K-means

Ward's Method

•**Ward's method** is one specific way to measure the "closeness" between two clusters during the agglomerative process. Instead of using simple distance metrics like single or complete linkage, it minimizes the total **within-cluster variance** (sum of squared differences from the mean of the cluster).
•The key idea is to merge clusters in such a way that the increase in the **within-cluster variance** is minimized. This encourages the formation of compact, spherical clusters.

How Ward's Method Works:
•**Step 1**: Start with each data point as its own cluster. For each pair of clusters, compute the **increase in variance** that would result from merging them.
•**Step 2**: Merge the pair of clusters whose combination results in the smallest increase in the total within-cluster variance.
•**Step 3**: Repeat the process until all data points are in one cluster or a predetermined number of clusters is reached.

Ward's Method

The **increase in variance** is calculated as:

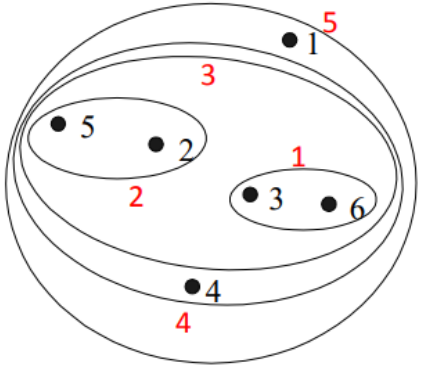
$$\text{Increase } \textit{in variance} = \frac{n_1 \cdot n_2}{n_1 + n_2} \cdot \| \mathbf{c}_1 - \mathbf{c}_2 \|^2$$

where:

- n_1 and n_2 are the sizes of the two clusters being merged,
- $\mathbf{c}_1$ and $\mathbf{c}_2$ are the centroids (means) of the two clusters,
- $\| \mathbf{c}_1 - \mathbf{c}_2 \|$ is the Euclidean distance between the centroids of the two clusters.

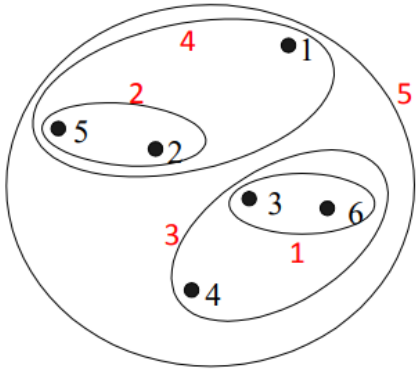
This formula reflects how much merging the two clusters would increase the variance (spread) of the resulting cluster.

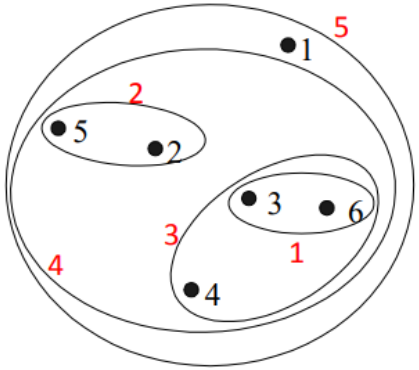
Comparison



MIN

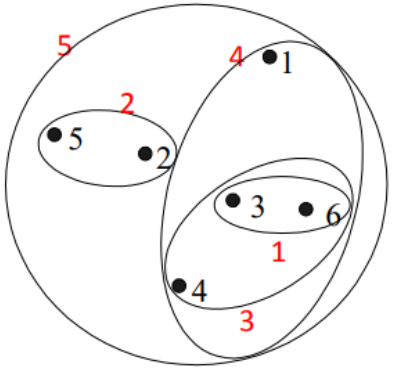
MAX





Group Average

Ward's Method



Time and Space Requirements

- $O(N^2)$ space since it uses the distance matrix
  - N is the number of points

- $O(N^3)$ time in many cases
  - There are N steps and at each step the size, N^2, distance matrix must be updated and searched
  - Complexity can be reduced to $O(N^2 \log(N))$ time for some approaches

Strengths

- Do not have to assume any particular number of clusters
  - Any desired number of clusters can be obtained by 'cutting' the dendrogram at the proper level
- They may correspond to meaningful taxonomies
  - e.g., shopping websites—electronics (computer, camera, ..), furniture, groceries

Problems and Limitations

- Once a decision is made to combine two clusters, it cannot be undone
- No objective function is directly minimized
- Different schemes have problems with one or more of the following:
  - Sensitivity to noise and outliers
  - Difficulty handling different sized clusters and irregular shapes
  - Breaking large clusters

Take-away Message

- Agglomerative and divisive hierarchical clustering
- Several ways of defining inter-cluster distance
- The properties of clusters outputted by different approaches based on different inter-cluster distance definition
- Pros and cons of hierarchical clustering